Stochastic Calculus and Option Pricing
Lecture 10 Exercises

10 Week 10 — American Options and Optimal Stopping

Focus. This question extends optimal stopping to the writer's hedge and requires a policy argument rather than a binomial price computation.

Exercise 10.1 (A supermartingale gives a writer's hedge)**.** Work in a discrete-time, arbitrage-free market with money-market account B_t. Let G_t be the amount due if an American claim is exercised at date t. Suppose a self-financing portfolio has value X_t and X_t/B_t is a nonnegative supermartingale under a risk-neutral measure.

(a) Prove that if $X_t \geq G_t$ at every exercise date, then the initial portfolio value upper-bounds the expected discounted payment for every admissible exercise rule.

(b) Explain why checking only the terminal payoff G_T is insufficient for an American claim.

(c) What additional condition would make this superhedge cheapest among all portfolios that cover every exercise date? Relate your answer to the least supermartingale that dominates the discounted exercise payoff.